⚙□ Step 1: Install Dependencies

```
npm init -y
npm install express ejs pg body-parser
```

🗄□ Step 2: PostgreSQL Setup

Open PostgreSQL and run:

```
CREATE DATABASE notesdb;

\c notesdb

CREATE TABLE notes (
  id SERIAL PRIMARY KEY,
  content TEXT NOT NULL
);
```

🔌 Step 3: Database Connection (db.js)

```
const { Pool } = require("pg");

const pool = new Pool({
  user: "postgres",
  host: "localhost",
  database: "notesdb",
  password: "your_password",
  port: 5432,
});

module.exports = pool;
```

🚀 Step 4: Express Server (app.js)

```
const express = require("express");
const bodyParser = require("body-parser");
const pool = require("./db");

const app = express();

app.set("view engine", "ejs");
app.use(express.static("public"));
app.use(bodyParser.urlencoded({ extended: true }));
```

```
// Home route
app.get("/", async (req, res) => {
  const result = await pool.query("SELECT * FROM notes ORDER BY id DESC");
  res.render("index", { notes: result.rows });
});

// Add note
app.post("/add", async (req, res) => {
  const { content } = req.body;
  await pool.query("INSERT INTO notes (content) VALUES ($1)", [content]);
  res.redirect("/");
});

// Delete note
app.post("/delete/:id", async (req, res) => {
  const id = req.params.id;
  await pool.query("DELETE FROM notes WHERE id=$1", [id]);
  res.redirect("/");
});

app.listen(3000, () => {
  console.log("Server running on http://localhost:3000");
});
```

🎯 Step 5: Frontend (views/index.ejs)

```
<!DOCTYPE html>
<html>
<head>
  <title>Notes App</title>
  <link rel="stylesheet" href="/style.css">
</head>
<body>

  <h1>📝 Notes App</h1>

  <form action="/add" method="POST">
    <input type="text" name="content" placeholder="Write a note..." required>
    <button type="submit">Add</button>
  </form>

  <ul>
    <% notes.forEach(note => { %>
      <li>
        <%= note.content %>
        <form action="/delete/<%= note.id %>" method="POST"
style="display:inline;">
          <button>❌</button>
        </form>
      </li>
    <% }) %>
  </ul>
```

```
</body>
</html>
What is this?
```

🎨 Step 6: CSS (public/style.css)

```
body {
  font-family: Arial;
  text-align: center;
  background: #f5f5f5;
}

h1 {
  color: #333;
}

form {
  margin: 20px;
}

input {
  padding: 10px;
  width: 250px;
}

button {
  padding: 10px;
  cursor: pointer;
}

li {
  list-style: none;
  margin: 10px;
  padding: 10px;
  background: white;
  display: inline-block;
}
```

□ Step 7: Optional JS (public/script.js)

```
console.log("Notes App Loaded");
```

▶□ Run the Project

```
node app.js
```

Open:

```
http://localhost:3000
```

💡 How It Works (Simple)

- User writes a note → sends POST request
- Express receives it → saves to PostgreSQL
- Page reloads → shows all notes from DB
- Delete button removes note from DB

🚀 Next Improvements (if you want)

- Edit notes ✏️
- Add timestamps 🕒
- User login system 🔐
- Search notes 🔍
- Convert to React frontend ⚛️

If you want, I can:
👉 Turn this into a **PDF file**
👉 Add **authentication (login/signup)**
👉 Or upgrade it to a **React + API full project**